	Multi-cloud	Hybrid cloud
Definition	The use of multiple cloud platforms from different providers.	A combination of on-premises infrastructure with public cloud services.
Benefits	Vendor independence, cost optimisation, disaster recovery, flexibility	Flexibility, cost optimisation, compliance, legacy system support
Challenges	Complexity, inconsistent APIs and tools, data governance	Complexity, integration, security
Characteristics	Ideal for businesses seeking maximum flexibility and vendor independence.	Suitable for businesses that want to gradually migrate to the cloud or maintain control over certain workloads.
Use cases	Ideal for large enterprises with complex IT environments and high tolerance for risk.	Well-suited for businesses with a mix of legacy systems and modern applications.
Example	A company using AWS for development, Azure for production, and GCP for analytics.	A company running critical applications on-premises while using the cloud for non-critical workloads or peak demand.

Table 1: Multi-cloud vs hybrid cloud

Multi-cloud: This refers to the use of multiple cloud platforms from different providers, such as AWS, Azure, and GCP. It offers increased flexibility, vendor independence, and disaster recovery capabilities.

Hybrid cloud: A hybrid cloud combines on-premises infrastructure with public cloud services. This approach allows businesses to maintain control over certain workloads while leveraging the benefits of the cloud for others.

By carefully considering the factors listed in Table 1, businesses can make informed decisions about whether to adopt a multi-cloud or hybrid cloud strategy and leverage the benefits while mitigating the risks. Some additional considerations are listed below.

- ***Security***: Both multi-cloud and hybrid cloud environments require robust security measures to protect sensitive data and prevent unauthorised access.
- ***Governance***: Implementing effective governance frameworks is

	Open source software	Proprietary software
Cost	Typically, more affordable	Relatively more expensive
Flexibility	Highly customisable and adaptable	Limited flexibility and customisation
Community support	Strong community-driven support	Limited support from vendors
Interoperability	Typically designed to be interoperable with different platforms.	May be limited in interoperability.
Innovation	Promotes innovation through collaboration and experimentation.	Innovation is slower due to vendor control.
Vendor lock-in	Reduces the risk of vendor lock-in.	Can lead to vendor lock-in.
Control	Provides greater control over software.	Offers limited control over software.
Customisation	Allows for extensive customisation.	Customisation may be limited.
Security	Can be highly secure but requires careful management.	May have security vulnerabilities.
Licensing	Various licensing models available (e.g., GPL, Apache, MIT).	Typically licensed under proprietary terms.

Table 2: Open source vs proprietary containerisation software in multi-cloud and hybrid cloud environments